Fizika

9 sinf

MAVZU: Gaz molekulalarining harakat tezligi.

Gaz molekulalarining harakat tezligi

Harakatlanayotgan m_0 massali gaz molekulalalarining o‘rtacha kinetik energiyasi formulalari:

$$\overline{E_k} = \frac{m_0\overline{\vartheta^2}}{2} \qquad \overline{E_k} = \frac{3}{2}\cdot k \cdot \mathrm{T}$$

$$\frac{m_0\overline{\vartheta^2}}{2} = \frac{3}{2}\cdot \mathrm{k}\cdot \mathrm{T} \qquad \overline{\vartheta^2} = \frac{3\cdot k\cdot T}{m_0}$$

$$\overline{\vartheta} = \sqrt{\frac{3\cdot k\cdot T}{m_0}} \qquad \overline{\vartheta} = \sqrt{\frac{3\cdot k\cdot N_A\cdot T}{M}}$$

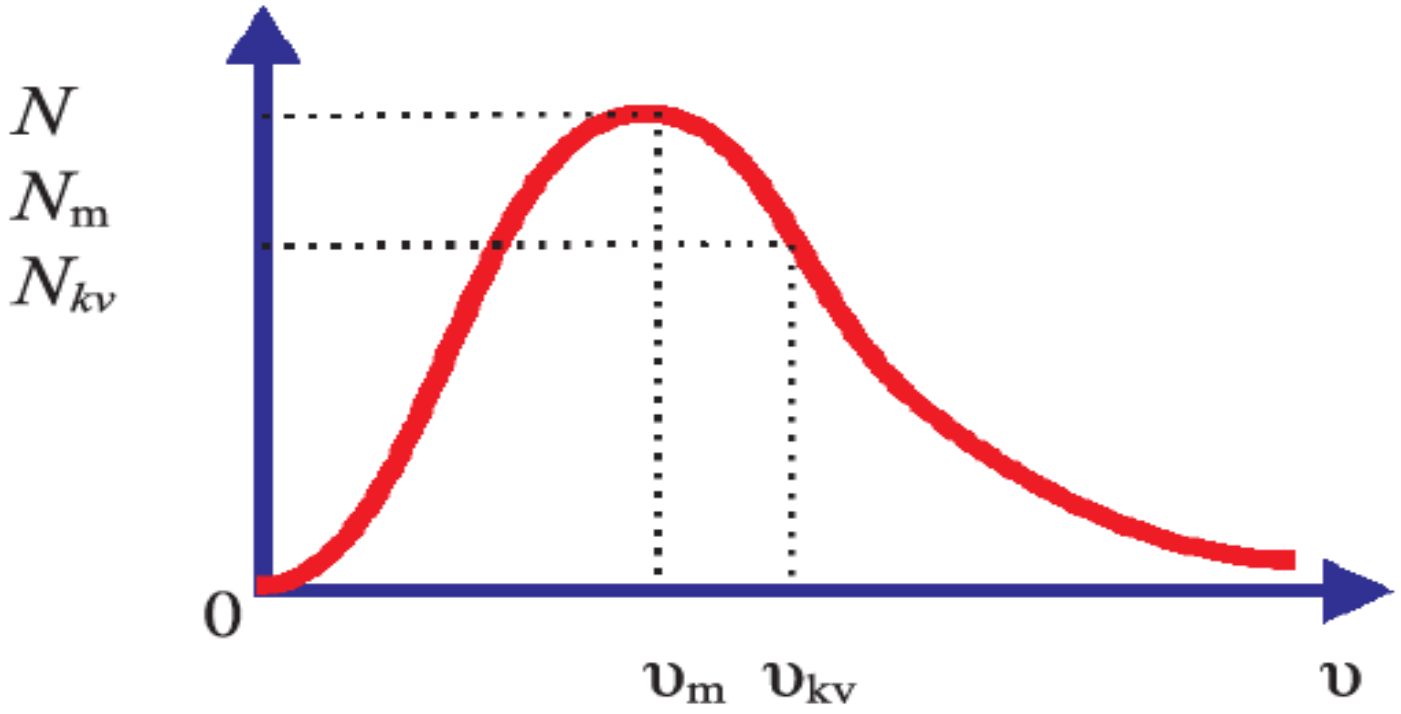
Bolsman doimiysi k bilan Avogadro doimiysi N_A ning ko‘paytmasiga universal (molyar) gaz doimiysi deb atash qabul qilingan.

$$\mathrm{R} = \mathrm{k} \cdot \mathrm{N_A} = 1{,}38 \cdot 10^{-23} \frac{J}{K} \cdot 6{,}02 \cdot 10^{23} \frac{1}{mol} = 8{,}31 \frac{J}{mol \cdot K}$$

$$\mathrm{R} = 8{,}31 \frac{J}{mol \cdot K}$$

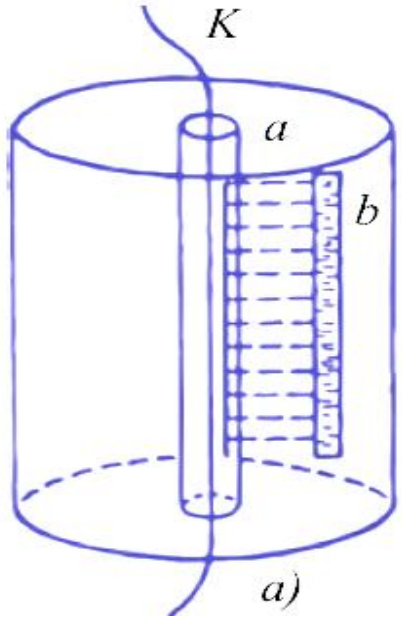
$$\overline{\vartheta} = \sqrt{\frac{3RT}{M}}$$

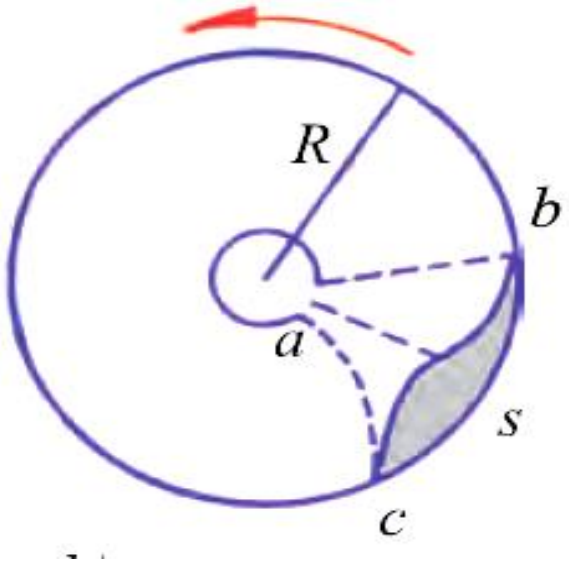
Ingliz fizigi J.Maksvell 1859- yilda nazariy yo‘l bilan gaz molekulalari turli tezliklar bilan harakatlanishini, ya’ni molekulalarning tezliklar bo‘yicha taqsimotini aniqladi.



Shtern tajribasi

Gaz molekulalarining tezligini 1920-yilda nemis fizigi Otto Shtern aniqlagan. $S = \omega R t$





Shtern tajribalarining natijalari

Shtern tajribasidan quyidagi natijalar kelib chiqadi:

1. Atomlar har xil tezlikda harakatlanadi;

2. Juda tez harakat qiladigan va juda sekin harakat qiladigan atomlar ozchilikni tashkil qiladi.

$$\vartheta_{o'rt} = \frac{R-r}{t} \qquad \vartheta_{o'rt} = \frac{\omega R(R-r)}{s}$$

Shtern tajribasi ideal gaz molekulyar-kinetik nazariyasining hamda Maksvellning gaz molekulalari tezliklari bo‘yicha taqsimoti haqidagi ta’limotining to‘g‘riligini tasdiqladi.

$$\overline{\vartheta}_{kumush} \approx 600\ m/s$$

$$\overline{\vartheta}_{vodorod} = 1838\ m/s$$

$$\overline{\vartheta}_{kislorod} \approx 640\ m/s$$

6-mashq 1-masala

Vodorod molekulasining -23℃ temperaturadagi o‘rtacha kvadratik tezligini aniqlang.

Berilgan:

$t = -23°C$

$M = 0{,}002 \frac{kg}{mol}$

$R = 8{,}31 \frac{J}{K \cdot mol}$

Formulasi:

$$\overline{\vartheta} = \sqrt{\frac{3RT}{M}}$$

Yechish:

$$T = 273 + (-27°C) = 250\ K$$

$$\overline{\vartheta} = \sqrt{\frac{3 \cdot 8{,}31 \cdot \frac{J}{K \cdot mol} \cdot 250\ K}{0{,}002\ kg/mol}} \approx$$

$$\approx 1765\ m/s$$

T.K: $\overline{\vartheta} - ?$

Javob: $\overline{\vartheta} \approx 1765\ m/s$

6-mashq 2-masala.

Qanday temperaturada kislorod molekulasining o‘rtacha kvadratik tezligi 500 m/s ga teng bo‘ladi?

Berilgan: O_2

$\overline{\vartheta}$ =500 m/s

M =0,032 kg/mol

$R = 8{,}31\ \frac{J}{K \cdot mol}$

Topish kerak: $t-?$

Formula:

$$\overline{\vartheta} = \sqrt{\frac{3 \cdot R \cdot T}{M}}$$

$$\overline{\vartheta^2} = \frac{3 \cdot R \cdot T}{M} \Rightarrow \mathrm{T} = \frac{\overline{\vartheta^2} \cdot M}{3R}$$

$$t = \mathrm{T} - 273$$

Yechish:

$$\mathrm{T} = \frac{(500\frac{m}{s})^2 \cdot 0{,}032\frac{kg}{mol}}{3 \cdot 8{,}31\ J/K \cdot mol} = 321\ \mathrm{K}$$

$$t = 321\ \mathrm{K} - 273 = 48℃$$

Javob: $t = 48℃$

Mustaqil bajarish uchun topshiriqlar

1. Havoda kislorod va vodorod molekulalarining o‘rtacha kvadratik tezliklari qanday farq qiladi?

2. Shtern tajribasidan qanday natijalar kelib chiqadi?

3. Gazning absolyut temperaturasi ikki marta ortganda undagi molekulalarning o‘rtacha kinetik energiyasi qanday o‘zgaradi?

4. Gazning absolyut temperaturasi to‘rt marta ortganda undagi molekulalarning o‘rtacha kvadratik tezligi qanday o‘zgaradi?